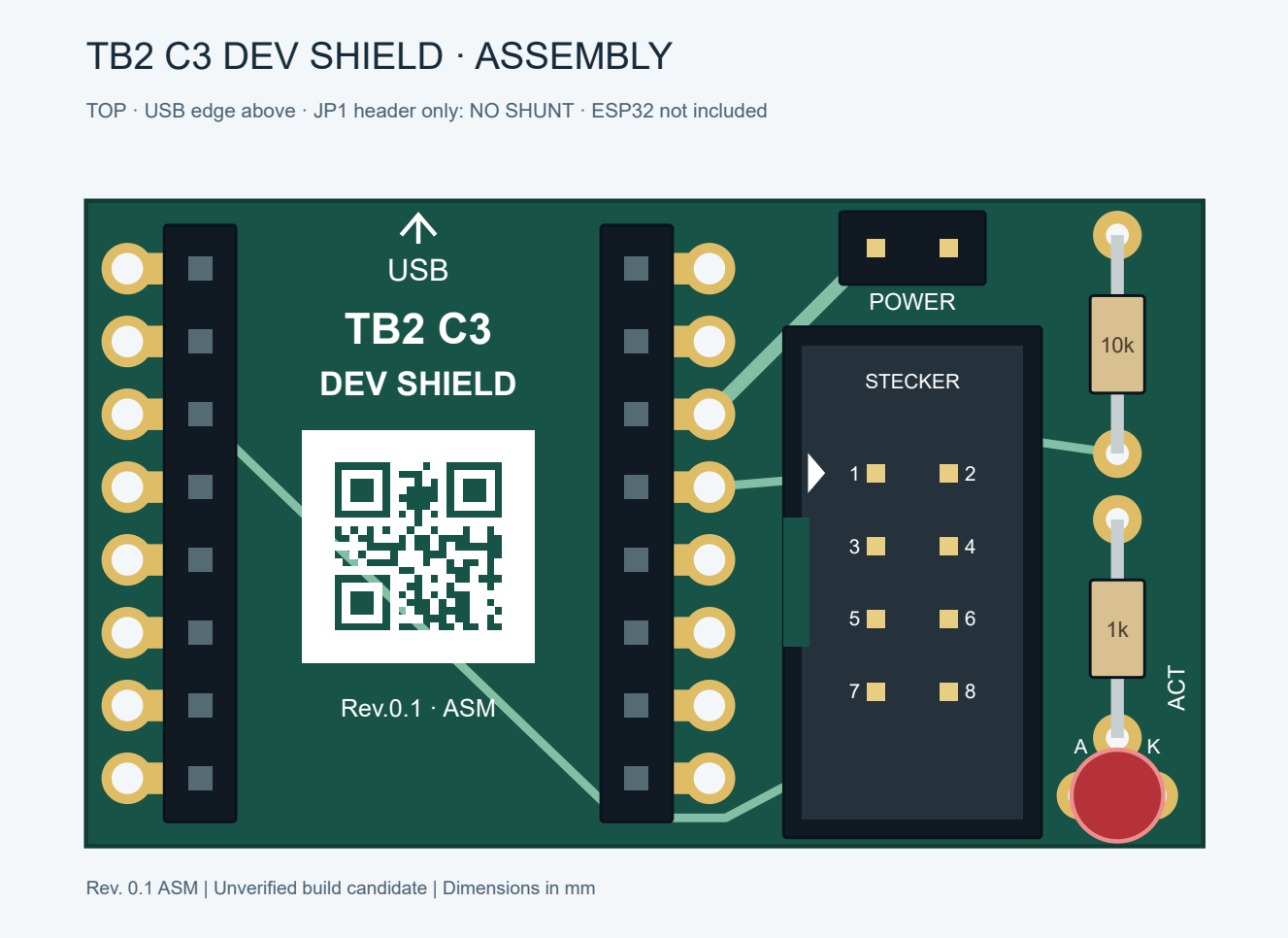
TB2 C3 DEV SHIELD · REV. 0.1 ASM

Assembly instructions

39 × 22.52 mm · 2 layers · all components on TOP · 7 THT parts · 32 solder joints · 0 SMT parts



JP1: fit the two-pin header only. Do not supply or fit a removable shunt. Leave the power header open. The ESP32 module, its male headers and the SOP8 cable are not included.

- **JL/JR:** two 1×8 female sockets in the INNER rows, opening upward; row spacing 15.24 mm. Leave all 16 outer alternative pads empty.
- **J1:** straight vertical 2×4 shrouded connector, opening upward, notch LEFT, pin 1 at the USB end. Do not rotate 180° or substitute a right-angle connector.
- **D1:** red LED, anode LEFT, cathode RIGHT toward the outer board edge. Match the cathode/flat side to the datasheet.
- **R1/R2:** horizontal axial parts, lead pitch 7.62 mm. R1 = 10 kΩ; R2 = 1 kΩ. No polarity.

Top view: USB edge at the top. No components on the bottom or underneath the ESP32. Name, USB arrow and QR are printed on both sides, readable from each side. Both QR codes encode the literal text Netfreak; keep them free of stickers. Unverified build candidate until physical acceptance.

PROCUREMENT AND INSPECTION

Parts per finished shield

Ref.	Qty	Reference part	Value / package
JL,JR	2	Wurth Elektronik 61300811821	2.54 mm; 8.5 mm body height; THT; body 20.82 x 2.5 mm
J1	1	Wurth Elektronik 61200821621	2.54 mm; THT; body 9.0 x 17.82 mm
JP1	1	Wurth Elektronik 61300211121	2.54 mm; 6 mm mating pins; THT; 1.10 mm PCB holes
R1	1	Yageo MFR25SFTE52-10K	10 kohm; 1%; 0.25 W; THT axial; form leads to 7.62 mm; body max 3.7 x 2.1 mm
R2	1	Yageo MFR25SFTE52-1K	1 kohm; 1%; 0.25 W; THT axial; form leads to 7.62 mm; body max 3.7 x 2.1 mm
D1	1	Kingbright WP710A10LID	3 mm red; low-current indicator; THT; 2.54 mm pitch

Resistor equivalents are allowed only if value, tolerance, power and maximum body dimensions match. Connector, socket and LED substitutions require approval. The reference resistor body is at most 3.7 × 2.1 mm; form leads to the 7.62 mm footprint without stressing the body.

Physical THT positions

Origin: **top-left corner in TOP view**, USB edge above. X increases rightward; Y increases downward. Millimetres. These are physical body/pin-group centres, not SMT machine coordinates.

Ref.	X [mm]	Y [mm]	Joints
JL	3.98	11.26	8
JR	19.22	11.26	8
J1	28.85	13.31	8
JP1	28.85	1.65	2
R1	36.0	5.01	2
R2	36.0	14.93	2
D1	36.0	20.75	2

The centres of JL/JR use only the populated inner rows. The unpopulated duplicate pads are not included in those centres. Orientation is defined by the drawing and notes; do not infer it from a generic pick-and-place rotation.

Electrical and mechanical inspection

1. Inspect solder joints and component polarity. Ensure the two socket rows are parallel and fit a 15.24 mm mating fixture without force.
2. Check that JP1 is fitted but open. There must be no direct copper/short connection between its two pins. Do not solder a permanent bridge.
3. Check the signal mapping below and inspect for adjacent-pin shorts. R1 and the LED branch are intentional component paths, not copper shorts.
4. Deliver the requested number of separate assembled shields. Confirm panelization and depanelization in the quote. No firmware programming, write/erase test, ESP32 or shunt procurement.

```
ESP32 GPIO4 → J1.1 → SOP8.1 CS     ESP32 GPIO5 → J1.3 → SOP8.2 MISO
ESP32 GPIO6 → J1.6 → SOP8.6 CLK    ESP32 GPIO7 → J1.8 → SOP8.5 MOSI
ESP32 GND   → J1.7 → SOP8.4 GND   Target 3V3 → J1.2 / J1.5 / J1.4
ESP32 3V3 → JP1.1   JP1.2 → Target 3V3
Target 3V3 → R1 10k → CS     Target 3V3 → R2 1k → D1.A; D1.K → CS
```

The user supplies a shunt later to enable target power. JP1 does not prevent SPI backfeed. Supply/load, cable fit, LED operation and two full flash reads remain physical acceptance checks.